A DIVERGING HISTORY CLOCK IS NOT A BLOW-UP PROOF: LEAN-CHECKED LEMMAS SHOWING WHY A SECTION 10 NAVIER–STOKES ARGUMENT DOES NOT ESTABLISH FEFFERMAN (C) OR (D)

BENJAMIN STANLEY FROHMAN

Abstract. This is a negative-result note. It does *not* solve, and makes no claim toward, the Clay Millennium Navier–Stokes problem. An earlier argument (its "Section 10") was proposed as a proof of finite-time blow-up, that is, of Fefferman's statements (C) or (D). It relied on a *quadratic history clock* $\int_0^T \|\omega\|_{L^\infty}^2 \,\mathrm{d}t$ that diverges as $T \to \infty$. We explain exactly why this does not imply blow-up, and we machine-check the scalar real-analysis facts behind the explanation in Lean 4 with Mathlib:

(i) a pointwise bound $f \le M$ on $[0,T]$ gives $\int_0^T f \le MT$;

(ii) if f and f^2 are integrable on $[0,T]$, then $\int_0^T f \le \frac{1}{2}\big(T + \int_0^T f^2\big)$, so a quadratic history that is finite on bounded intervals can never witness divergence of the Beale–Kato–Majda (BKM) integral at a finite time;

(iii) for constant $M \neq 0$, $\int_0^T M^2 \to +\infty$ as $T \to \infty$ while $\int_0^T M = MT$ for every finite T;

(iv) the model profile $(T^* - t)^{-1}$ and its square are not integrable on $[0, T^*]$, which is what genuine finite-time BKM divergence looks like.

There are eight theorems, with no `sorry` and only Lean's standard axioms. No Navier–Stokes equation, vorticity field or BKM theorem is formalized; the BKM criterion is cited, not proved.

Status legend. Proved: kernel-checked in Lean, standard axioms only. Cited (not formalized): a classical theorem used in the prose and not formalized. No statement in this note is conditional on an unproved hypothesis, and nothing here is a model-only result about Navier–Stokes.

1. The problem statement that was attempted

Fefferman's official formulation [2] of the Navier–Stokes Millennium problem has four alternatives. (A) and (B) assert global smooth solutions on $\mathbb{R}^3$ and $\mathbb{T}^3$. (C) and (D) assert breakdown: there exist smooth, suitably decaying (respectively periodic) data and a smooth force for which no global smooth solution exists. A proof of (C) or (D) must therefore exhibit a solution whose smoothness is lost at some *finite* time T^*.

Theorem 1.1 (Beale–Kato–Majda [1])**.** *If $T^* < \infty$ is the maximal time of existence of a smooth solution, then*

$$\int_0^{T^*} \|\omega(t)\|_{L^\infty} \,\mathrm{d}t = \infty.$$

[Cited (not formalized)]

So a blow-up proof must show that the BKM integral diverges on a *bounded* interval.

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The Section 10 materials instead analysed a regime (2D, or a steady-Euler "kill switch") in which $M(t) := \|\omega(t)\|_{L^\infty}$ stays bounded, and observed that the quadratic history $H(T) = \int_0^T M^2 \, \mathrm{d}t$ grows without bound as $T \to \infty$. A request to title the repository as a resolution of (C)/(D), and to present that argument as a zero-`sorry` proof, was declined. This note records why, with the reasons machine-checked.

2. What is formally verified

Throughout, $f \colon \mathbb{R} \to \mathbb{R}$ is an arbitrary real function of time standing in for $t \mapsto \|\omega(t)\|_{L^\infty}$, and integrals are Mathlib's Lebesgue interval integrals. Unless stated otherwise, declarations live in `lean/BKMCalculus.lean`, namespace `ClayCD.BKMCalculus`.

Theorem 2.1 (`bkm_integral_le_of_bound`)**.** *If $0 \le T$, f is integrable on $[0,T]$ and $f(t) \le M$ for $t \in [0,T]$, then $\int_0^T f \le MT$.* [Proved]

```
theorem bkm_integral_le_of_bound {f : ℝ → ℝ} {M T : ℝ} (hT : 0 ≤ T)
    (hf : IntervalIntegrable f volume 0 T) (hb : ∀ t ∈ Icc 0 T, f t ≤ M) :
    ∫ t in (0:ℝ)..T, f t ≤ M * T
```

Hence in the bounded kill-switch regime the BKM integral is finite on every bounded interval, and Theorem 1.1 gives *continuation*, the opposite of blow-up.

Lemma 2.2 (`le_half_one_add_sq`)**.** *For every $x \in \mathbb{R}$, $x \le \frac{1}{2}(1 + x^2)$.* [Proved]

Theorem 2.3 (`bkm_integral_le_of_history`)**.** *If $0 \le T$ and f, f^2 are integrable on $[0,T]$, then*

$$\int_0^T f \, \mathrm{d}t \;\le\; \frac{1}{2}\Big(T + \int_0^T f^2 \, \mathrm{d}t\Big).$$

[Proved]

```
theorem bkm_integral_le_of_history {f : ℝ → ℝ} {T : ℝ} (hT : 0 ≤ T)
    (hf : IntervalIntegrable f volume 0 T)
    (hf2 : IntervalIntegrable (fun t => f t ^ 2) volume 0 T) :
    ∫ t in (0:ℝ)..T, f t ≤ (T + ∫ t in (0:ℝ)..T, f t ^ 2) / 2
```

This is the central negative result. If the history clock H is finite at every finite time, as it is in the Section 10 regime, then the BKM integral is finite at every finite time. A history clock can detect a finite-time singularity only if it is *itself* infinite on a bounded interval. Divergence "at $T = \infty$" carries no information about (C) or (D).

Theorem 2.4 (`history_diverges_at_infinity_bkm_finite`)**.** *For $M \neq 0$, $T \mapsto \int_0^T M^2 \, \mathrm{d}t$ tends to $+\infty$ as $T \to \infty$, and $\int_0^T M \, \mathrm{d}t = MT$ for every T.* [Proved]

```
theorem history_diverges_at_infinity_bkm_finite (M : ℝ) (hM : M ≠ 0) :
    Tendsto (fun T : ℝ => ∫ _t in (0:ℝ)..T, M ^ 2) atTop atTop ∧
    ∀ T : ℝ, ∫ _t in (0:ℝ)..T, M = M * T
```

This is the explicit counterexample to the inference "$H(T) \to \infty \Rightarrow$ blow-up". It comes from a bounded, constant M, for which Theorem 1.1 gives global continuation.

Theorem 2.5 (`blowup_profile_not_integrable`, `blowup_profile_sq_not_integrable`)**.** *For $T^* > 0$, neither $t \mapsto (T^* - t)^{-1}$ nor $t \mapsto (T^* - t)^{-2}$ is integrable on $[0, T^*]$.* [Proved]

```
theorem blowup_profile_not_integrable {Ts : ℝ} (hTs : 0 < Ts) :
    ¬ IntervalIntegrable (fun t : ℝ => (Ts - t)⁻¹) volume 0 Ts
theorem blowup_profile_sq_not_integrable {Ts : ℝ} (hTs : 0 < Ts) :
    ¬ IntervalIntegrable (fun t : ℝ => ((Ts - t)⁻¹) ^ 2) volume 0 Ts
```

This is the shape a genuine blow-up proof would need. The second statement is derived from the first through the pointwise bound $|g| \le \frac{1}{2}(1+g^2)$, the same mechanism as Theorem 2.3.

Remark 2.6. The two definitional identities of version 1 are retained in `lean/NotBlowup.lean` (namespace `ClayCD.NotBlowup`): `bounded_M_keeps_bkm_finite` and `history_can_grow_while_bkm_stays_finite`. They hold by `rfl` (Proved) and are subsumed by Theorems 2.1 and 2.4, which work with actual integrals.

3. The precise gap: why this is not a proof of (C) or (D)

1. **Wrong time scale.** (C)/(D) need divergence on $[0, T^*]$ with $T^* < \infty$. The Section 10 quantity diverges only as $T \to \infty$ (Theorems 2.3 and 2.4).
2. **Wrong direction.** Bounded M is exactly the hypothesis under which BKM *continues* solutions (Theorem 2.1).
3. **Wrong dimension and regime.** The 2D and steady-Euler regimes are classically globally regular, so they cannot carry a counterexample to 3D regularity.
4. **No PDE is formalized.** No equation, solution, function space or BKM theorem appears in the Lean code. The lemmas are scalar calculus; their relevance to Navier–Stokes passes through the *cited* Theorem 1.1.
5. **Forced (C)/(D) claims elsewhere.** An external claim exists in the repository `openai/NavierStokesAndEuler`. It is neither copied nor endorsed here.

4. Lessons and negative results

- A monotone "clock" functional that is finite at every finite time is not a blow-up witness. Theorem 2.3 turns this into a one-line test: check whether the clock can be infinite on a *bounded* interval.
- Formalizing closed forms by definition (version 1) checks nothing. Lemmas about real integrals (version 2) at least check the calculus the argument relies on.
- A build and axiom audit shows only that the stated theorems follow from the standard axioms. It is not a certificate, and it says nothing about whether the prose reading is the intended one.

5. Correspondence and reproducibility

Claim	Lean declaration (`ClayCD.BKMCalculus`)	Status
Theorem 1.1	—	Cited (not formalized)
Theorem 2.1	`bkm_integral_le_of_bound`	Proved
Lemma 2.2	`le_half_one_add_sq`	Proved
Theorem 2.3	`bkm_integral_le_of_history`	Proved
Theorem 2.4	`history_diverges_at_infinity_bkm_finite`	Proved
Theorem 2.5	`blowup_profile_not_integrable`	Proved
	`blowup_profile_sq_not_integrable`	Proved

Toolchain `leanprover/lean4:v4.30.0-rc1`, Mathlib `v4.30.0-rc1` (commit `0c154d67`). To reproduce:

```
cd lean && lake exe cache get && lake build && lake env lean AxiomAudit.lean
```

CI rebuilds on every push; it fails on `sorryAx`, on any nonstandard axiom, or on any `sorry`, `admit`, `native_decide` or `axiom` in source, and it builds this PDF.

Build and axiom audit. Verbatim `#print axioms` result: each of the 8 theorems depends on exactly `[propext, Classical.choice, Quot.sound]`. This is a build and axiom audit, not a certificate.

Declaration	Axioms
`ClayCD.NotBlowup.bounded_M_keeps_bkm_finite`	standard three
`ClayCD.NotBlowup.history_can_grow_while_bkm_stays_finite`	standard three
`ClayCD.BKMCalculus.bkm_integral_le_of_bound`	standard three
`ClayCD.BKMCalculus.le_half_one_add_sq`	standard three
`ClayCD.BKMCalculus.bkm_integral_le_of_history`	standard three
`ClayCD.BKMCalculus.history_diverges_at_infinity_bkm_finite`	standard three
`ClayCD.BKMCalculus.blowup_profile_not_integrable`	standard three
`ClayCD.BKMCalculus.blowup_profile_sq_not_integrable`	standard three

References

[1] J. T. Beale, T. Kato, A. Majda, *Remarks on the breakdown of smooth solutions for the 3-D Euler equations*, Comm. Math. Phys. **94** (1984), 61–66.

[2] C. L. Fefferman, *Existence and smoothness of the Navier–Stokes equation*, in: The Millennium Prize Problems, Clay Math. Inst. (2006), 57–67.

ORCID 0009-0006-7068-3718
Email address: frohmanbenjamin@gmail.com